

Forecasting Near-Term Benzene Concentration in Urban Air

Executive Summary - Predictive Analytics Midterm Project - Zaky Hafiedz - 24130500006

Problem Background

Benzene (C₆H₆) is a regulated, carcinogenic air pollutant tied primarily to vehicle traffic and fuel emissions. Reference-grade instruments for measuring it are expensive, which motivated the original research behind this dataset: testing whether a low-cost multisensor device could estimate benzene levels reliably. Building on that idea, this project asks a related, practical question: can next-hour benzene concentration be forecast accurately enough from recent sensor history to support an early-warning system for an environmental monitoring agency? A reliable short-term forecast would let such an agency issue alerts before pollution peaks, rather than only reporting them after the fact.

Dataset and Target

Data: UCI Machine Learning Repository, "Air Quality" dataset (id 360), donated by S. De Vito. Source paper: De Vito et al. (2008), "On field calibration of an electronic nose for benzene estimation in an urban pollution monitoring scenario," *Sensors and Actuators B: Chemical*, 129(2), 750–757.

Scope: 9,358 hourly readings from a chemical multisensor device mounted at road level in a heavily trafficked area of an Italian city, spanning March 2004 to April 2005 (about 13 months that is slightly longer than the summary description on the source page implies).

Unit of analysis: one hourly-averaged reading of sensor and reference-analyzer values.

Target: C₆H₆(GT), the reference-analyzer benzene concentration (μg/m³), forecast one hour ahead.

Key limitation: missing readings are encoded as the placeholder value -200 rather than left blank, and one column (NMHC(GT)) is missing for the large majority of the deployment period due to reference-sensor failure. Both issues required explicit handling before any analysis (see Methods).

Methods Compared

- **Data preparation:** Interpolation over deletion for missing values, since the forecasting target relies on consistent hourly spacing, missing Benzene readings were filled using time-based interpolation rather than dropped, to avoid introducing artificial gaps into lag and rolling features.
- **Feature engineering:** Lag features (1h, 24h, 168h), rolling mean/std (3h, 6h, 24h windows), and cyclical hour/day/month encodings, chosen based on the clear daily and weekly traffic patterns and the seasonal dip found during exploratory analysis.
- **Naïve baseline:** Predicting next-hour value using the current value. This is a strong benchmark for this series given its hour-to-hour autocorrelation, not a token comparison.
- **SARIMA:** A Seasonal ARIMA model with daily seasonality (order (2,1,2), seasonal order (1,1,1,24)), representing the classical statistical forecasting approach.
- **Random Forest Regressor:** Trained on all engineered features, representing the machine-learning forecasting approach and the only model evaluated for feature importance.

All models were evaluated on an identical, chronologically held-out test set (final 20% of the timeline, no shuffling) to prevent information from the future leaking into training, and all preprocessing (scaling) was fit on training data only.

Main Results

Model	MAE (μg/m ³)	RMSE (μg/m ³)	MAPE (%)
Naive Baseline	2.20	3.67	31.19%
SARIMA	7.37	9.27	147.25%
Random Forest (selected)	1.80	2.74	31.43%

Random Forest was the only model to outperform the naive baseline, improving MAE by 18.24%. SARIMA performed considerably worse than both, not because statistical forecasting is unsuited to this problem, but because it was evaluated as a

single long-horizon static forecast across the entire 1,872-hour test period with no access to real observed values along the way, causing it to revert toward the series average and lose all spike structure. A fairer, rolling one-step-ahead SARIMA evaluation was identified as future work rather than repeated here, to preserve a stable, already-verified result.

Feature importance analysis of the selected Random Forest model found that a single raw sensor reading (PT08.S2(NMHC), correlated at 0.982 with benzene) and the current-hour benzene reading itself together account for roughly three-quarters of the model's predictive power. The purpose-built lag, rolling, and time features each contribute a smaller individual share. Error analysis further showed a mild systematic bias, with the model's largest errors occurring specifically during sharp pollution spikes, which it consistently under-predicted.

Selected Model

Random Forest is recommended as the final model. It is the only approach that meaningfully beat a genuinely strong naive baseline, and it provides interpretable feature importances that a monitoring agency can act on directly (for example, prioritizing the reliability of the one sensor the model depends on most). SARIMA is not recommended as evaluated, though it remains worth revisiting with a rolling forecast setup in future work.

Recommendations

- **Operational use:** Deploy the Random Forest model for 1-hour-ahead benzene forecasting as a decision-support tool, not a standalone alert trigger.
- **Sensor prioritization:** Since roughly half of the model's predictive power comes from one sensor (PT08.S2(NMHC)), prioritize monitoring that sensor's calibration and uptime above other inputs.
- **Caution during spikes:** Treat forecasts issued during rapidly rising conditions with extra caution, since accuracy is weakest exactly during the high-pollution spikes an early-warning system is meant to catch.
- **Next steps:** Retrain periodically to capture seasonal and traffic-pattern drift, and re-evaluate SARIMA using a rolling one-step-ahead forecast for a fairer statistical comparison.

Limitations

- **Geographic scope:** Trained on one Italian city; would require retraining before use elsewhere.
- **Sensor dependence:** A large share of model accuracy depends on one specific low-cost sensor reading, a meaningful single point of failure; metal-oxide sensors are also documented to drift over time.
- **Data quality:** One raw dataset column (NMHC(GT)) had such extensive missingness it was effectively unusable and excluded from consideration.
- **Spike performance:** The model under-predicts during rare, sharp spikes that are precisely the conditions most relevant to public health alerting.
- **Evaluation asymmetry:** The reported SARIMA result reflects a disadvantageous evaluation setup and should not be read as evidence that statistical forecasting cannot work for this problem.
- **Forecast horizon:** Only 1-hour-ahead forecasting was evaluated; accuracy at longer horizons is unknown.